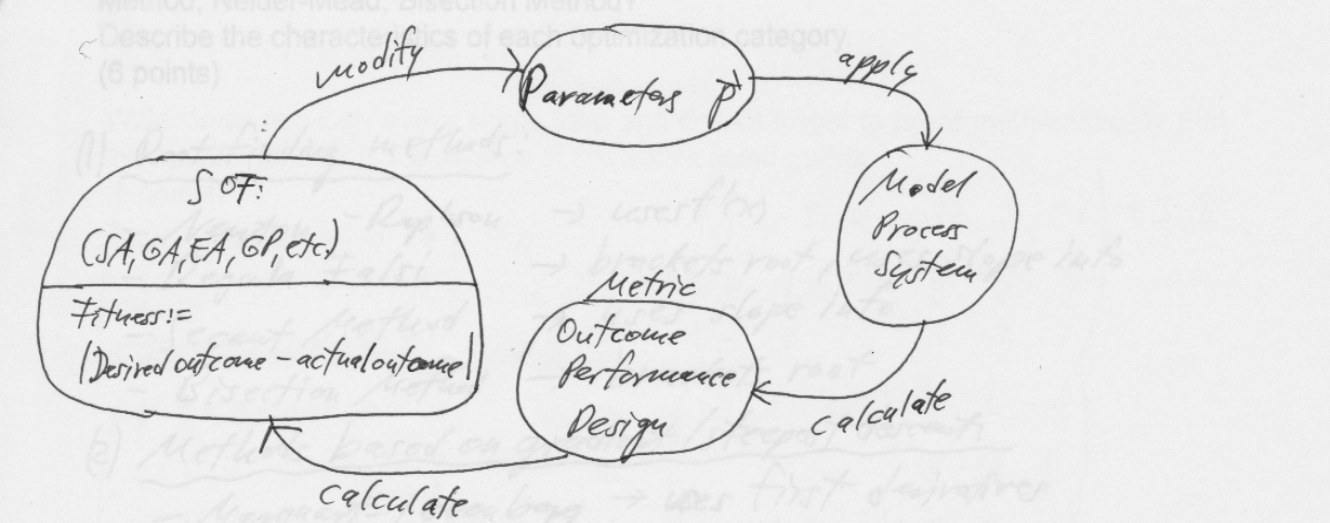
Questions for 2nd Midterm Exam ECE 330:

Name: _____

Question #1 (6 points total):

Describe the Stochastic Optimization Framework in detail (i.e., graphical sketch with labeling of all components and written explanation of how it works).
(6 points)



Explanations:

1) Initial parameter set $\vec{p}$
2) Parameters applied to model
3) Run model in forward fashion to calculate outcome
4) Calculate resulting fitness!=|desired outcome - actual outcome|
5) SOF uses this fitness to modify the parameters via algorithms such as SA, GA, EA, GP
6) new parameter set determined
7) Continue with step 2 until Fitness reaches user-defined target fitness

Question #2 (10 points total):

a) Describe in words the general procedure for Genetic Algorithms (GA).
(3 points)

b) Describe in words the general procedure for Simulated Annealing (SA).
(3 points)

c) List three genetic operators by name and point out which ones are not biologically motivated.
(4 points)

a) Similar to this:
- create a gene pool of random solutions (i.e., a set of random parameter vectors)

(*) – generate offspring/children by applying genetic operators (point mutation, crossover, inversion)
- evaluate fitness of offspring: if sufficient → STOP, FINISHED!
- replace some of the parents to maintain same overall number of genes, may apply Elitist approach to maintain fittest genes, also may introduce random genes occasionally to maintain diversity and gradient in gene pool
- loop back to (*) till exit criterion (i.e., target fitness reached or allowable number of iterations exceeded) reached

b) Similar to this:
- operates on single parameter vector that is chosen randomly

(*) – introduce random change within parameter vector (i.e., point mutation)
- evaluate fitness of modified parameter vector: if sufficient → STOP, FINISHED!
  - Accept change if lower energy/better fitness/better performance
  - Accept change if higher energy/worse fitness/worse performance only with certain Boltzmann-probability (enables tunneling)
  - Reject change otherwise
- loop back to (*) till exit criterion (i.e., target fitness reached or allowable number of iterations exceeded) reached

c) Point mutation (biologically motivated)
Crossover (biologically motivated)
Inversion (not biologically motivated)

Question #3 (8 points total):

The UA bookstore wants to liquidate 200 of its Wildcat shirts and 100 pairs of UA jogging pants from last season. They have decided to put together two offers, A and B. Offer A is a package of one Wildcat shirt and a pair of UA jogging pants (note: a pair of pants is ONE item), which will sell for $30. Offer B is a package of three Wildcat shirts and a pair of UA jogging pants, which will sell for $50. The UA bookstore does not want to sell less than 20 packages of Offer A and less than 10 of Offer B. How many packages of each do they have to sell to maximize the money generated from the promotion?
Solve this problem in detailed steps (description of steps, formulas, graphical sketches, actual calculation, etc.). Shortcuts will lead to point reductions!
(8 points)

1) define unknowns:

x = # of packages of offer A
y = # of packages of offer B

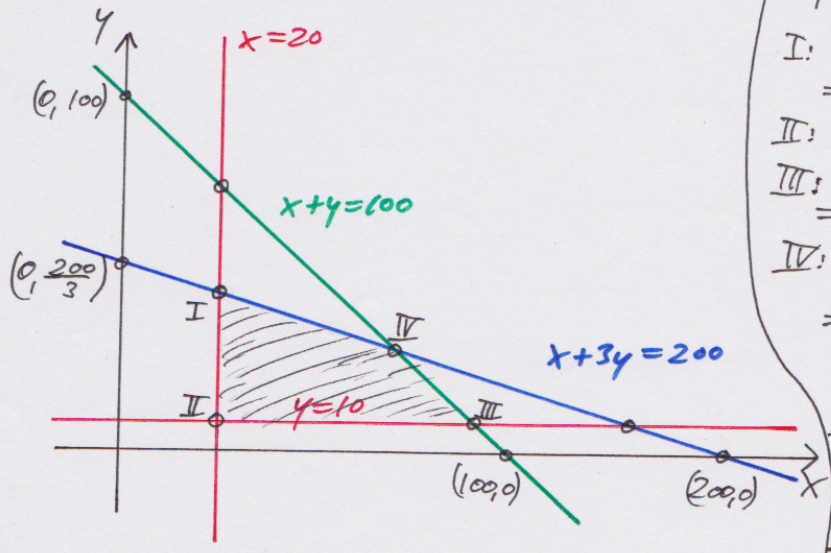
2) write objective function:

$f(x,y) = 30x + 50y$

3) write constraints as system of inequalities:

Shirts: $x + 3y \leq 200$
Pants: $x + y \leq 100$
$x \geq 20$
$y \geq 10$
and $x, y \in \mathbb{N}$

4) find set of feasible solutions:



5) Calculate coordinates of valid/feasible vertices:

I: $20+3y=200 \Rightarrow y=60$
$\Rightarrow (20,60)$
II: $(20,10)$
III: $x+10=100 \Rightarrow x=90$
$\Rightarrow (90,10)$
IV: $x+y=100 \Rightarrow y=100-x$
$x+3y=200$
$\Rightarrow x+300-3x=200$
$\Rightarrow x=50$
$\Rightarrow y=50$
$\Rightarrow (50,50)$

6) Calculate objective function value at each vertex:

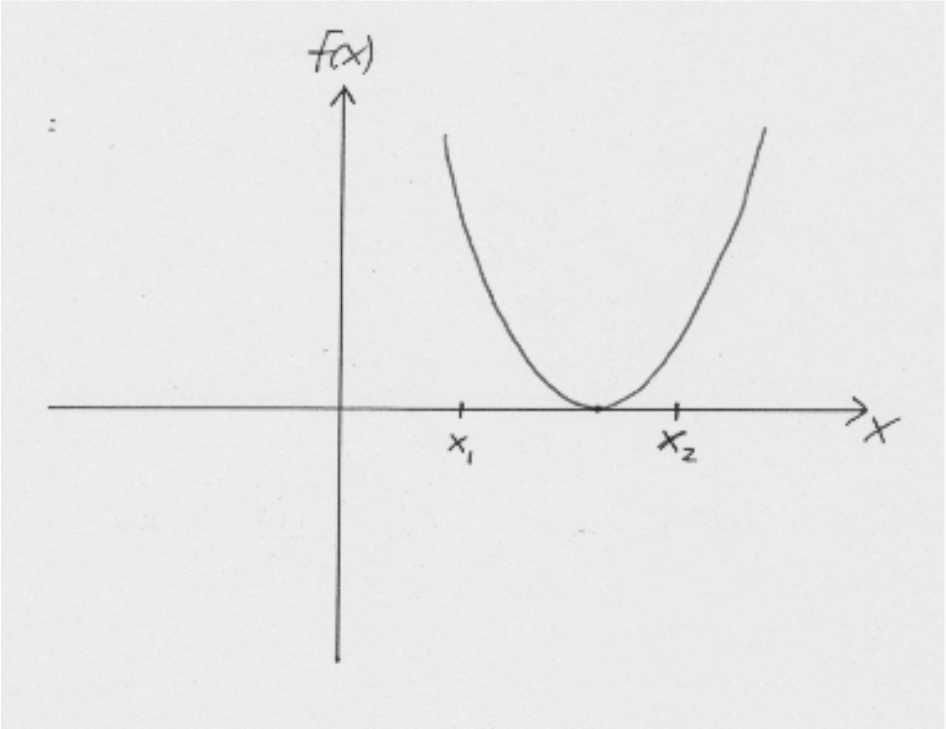
I: $f(20,60)=20\cdot30+60\cdot50=3600$
II: $f(20,10)=20\cdot30+10\cdot50=1100$

6) continued:

III: $f(90,10)=90\cdot30+10\cdot50=3200$
IV: $f(50,50)=50\cdot30+50\cdot50=4000$
$\Rightarrow$ answer: 50x offer A, 50x offer B yields maximum $

Question #4 (2 points total):

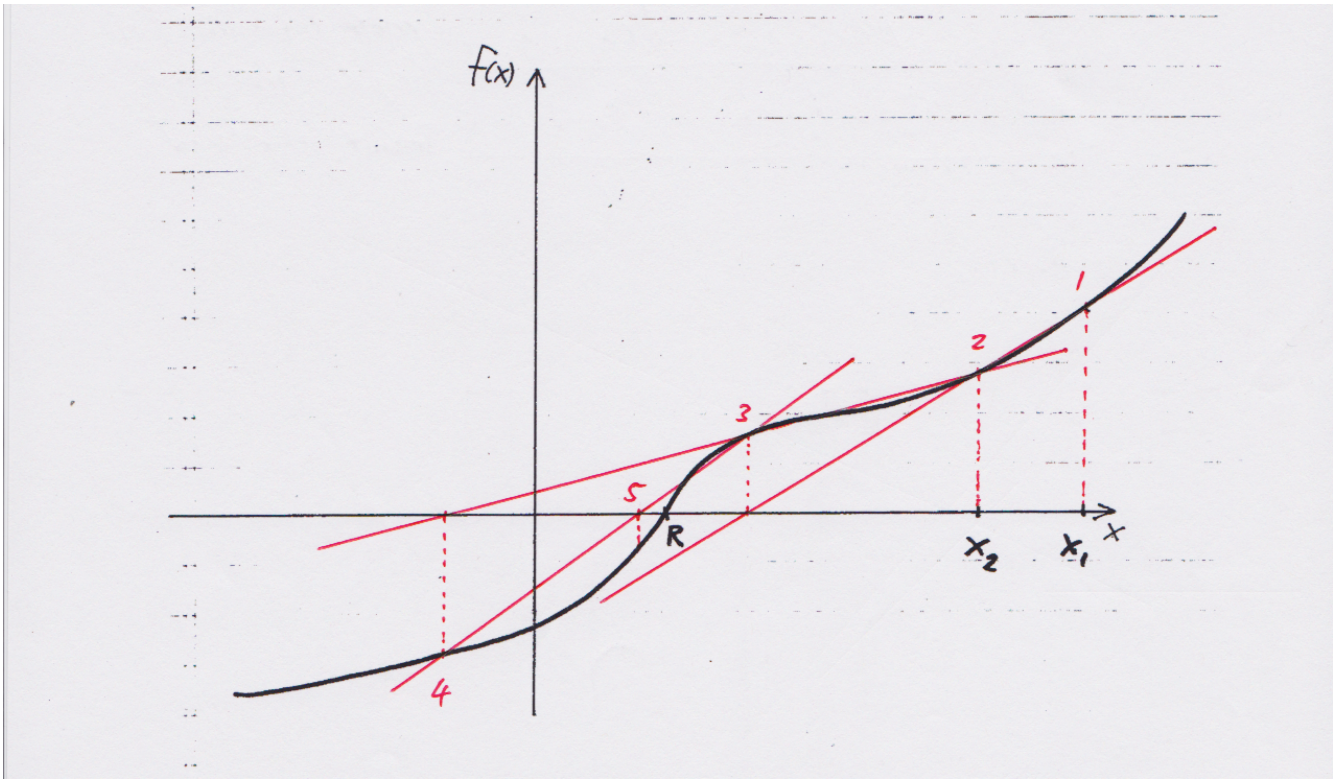
Draw 3 iterations (i.e., 3 updates) of the Bisection Method into the given graph with x1 and x2 as the two starting x-values.
(2 points)



Not applicable in this case, because the *Bisection Method* has to bracket the root at all times with one interval boundary function value being negative and the other one being positive. Here both function values $f(x_1)$ and $f(x_2)$ are positive, therefore the *Bisection Method* is NOT applicable in this case.

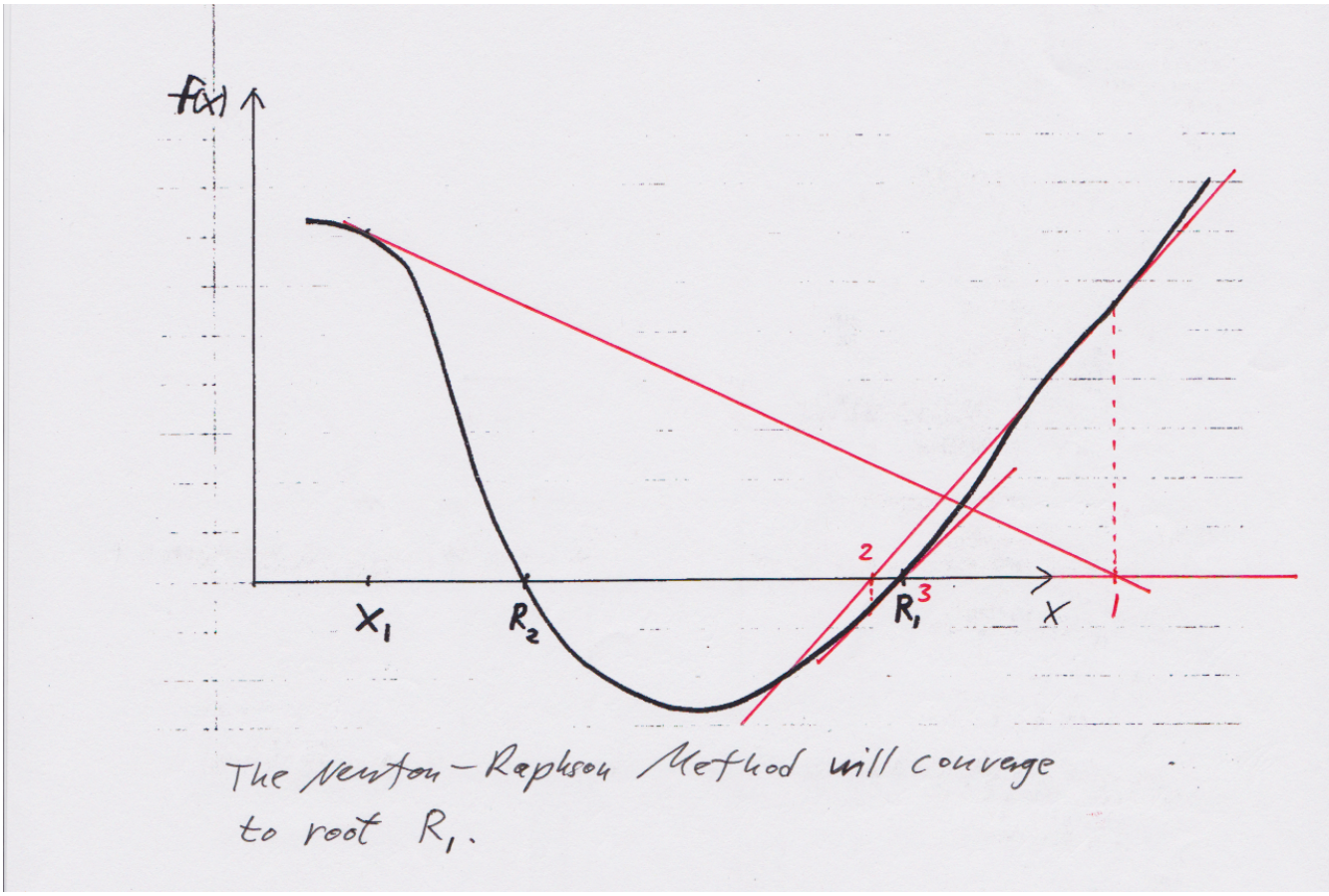
Question #5 (3 points total):

Draw 3 iterations (i.e., 3 updates) of the Secant Method into the given graph with x1 and x2 as the two starting x-values.
(3 points)



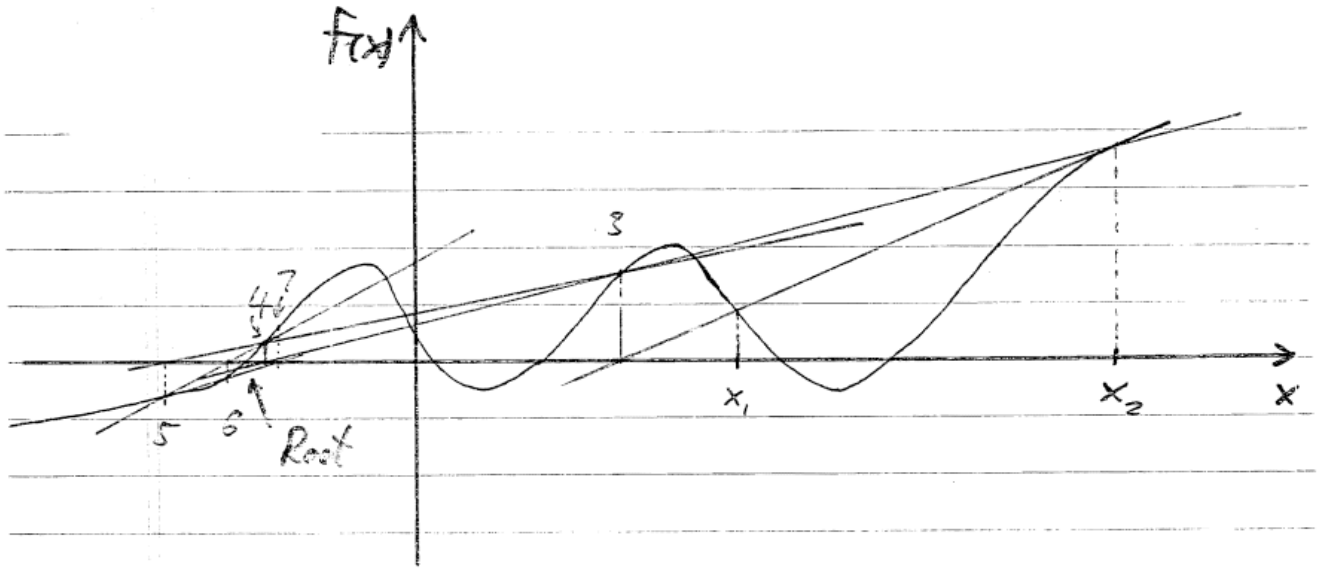
Question #6 (3 points total):

Using the 1D Newton-Raphson Method, which root will it converge to when using x1 as the starting x-value? Show several iterations to convincingly draw your conclusion. (3 points)



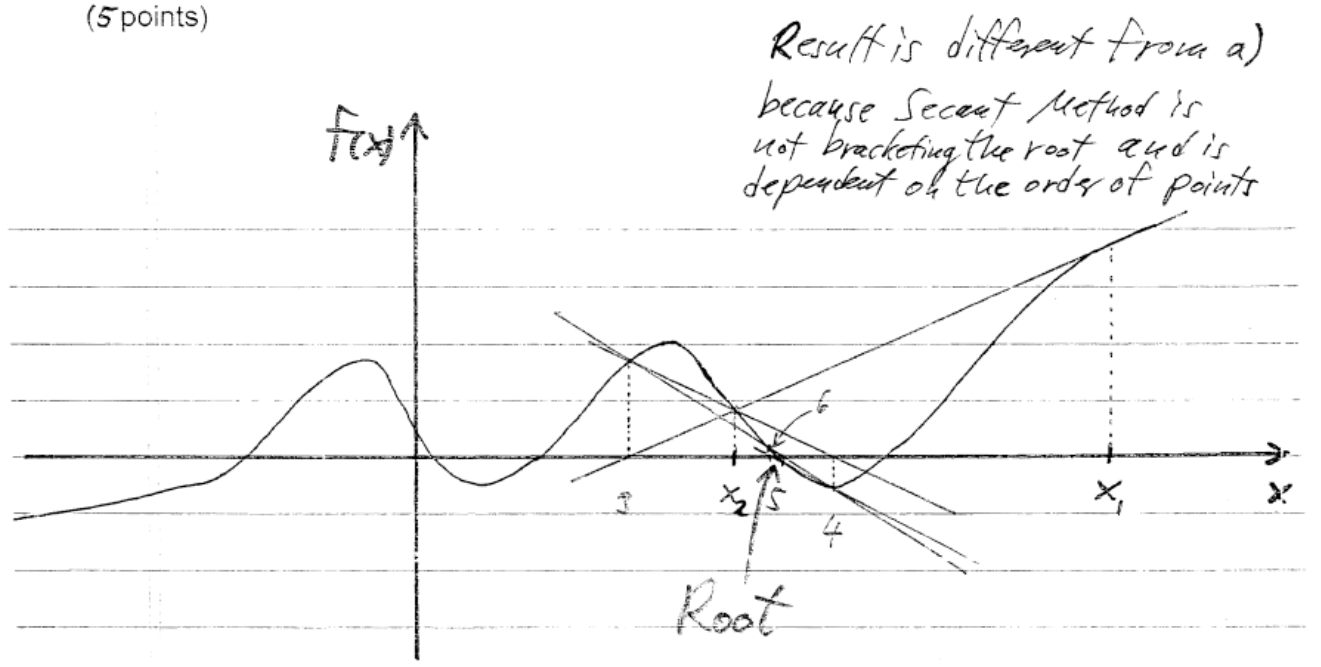
Question #7 (6 points total):

a) Applying the Secant Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b.
(3 points)

(5 points)



Question #8 (6 points total):

a) Derive mathematically from scratch the update formula for the general multi-dimensional Newton-Raphson Method.
(2 points)

b) Given $f(x,y) = xy^3$ and $g(x,y) = \sqrt{x} + y$ determine explicitly the parameter vector$(x_1, y_1)$of the next iteration using the Newton-Raphson Method in two dimensions given the start vector$(1,1)$.
(4 points)

a)

Method

We have N functional relations to be zeroed with variables x_i with $i=1,\dots,N$

$$F_i(x_1, x_2, \dots, x_N) = 0 \quad \forall i=1,\dots,N$$

$$\Rightarrow \vec{F}(\vec{x}) = 0$$

in neighborhood of $\vec{x}$ each function $F_i(\vec{x})$ can be expanded in Taylor series:

$$F_i(\vec{x}+\delta\vec{x}) = F_i(\vec{x}) + \underbrace{\sum_{j=1}^{N} \frac{\partial F_i(\vec{x})}{\partial x_j} \delta x_j}_{\text{matrix of partial derivatives of } F_i \text{ is Jacobian matrix } \mathcal{J}: J_{ij} = \frac{\partial F_i(\vec{x})}{\partial x_j}} + \mathcal{O}(\delta\vec{x}^2) \quad \forall i=1,\dots,N$$

$$\Rightarrow \vec{F}(\vec{x}+\delta\vec{x}) = \vec{F}(\vec{x}) + \mathcal{J}\,\delta\vec{x} + \mathcal{O}(\delta\vec{x}^2) \quad \text{(ignore error term)}$$

$$\vec{F}(\vec{x}+\delta\vec{x}) = 0$$

$$\Rightarrow 0 = \vec{F}(\vec{x}) + \mathcal{J}\,\delta\vec{x}$$

$$\Rightarrow \mathcal{J}\,\delta\vec{x} = -\vec{F}(\vec{x})$$

$$\Rightarrow \boxed{\delta\vec{x} = -\mathcal{J}^{-1}\vec{F}(\vec{x})}$$

$$\Rightarrow$$

Newton-Raphson Method in N Dim.

N update equations:

$$\vec{x}_{new} = \vec{x}_{old} + \delta\vec{x}$$

b) Jacobi matrix

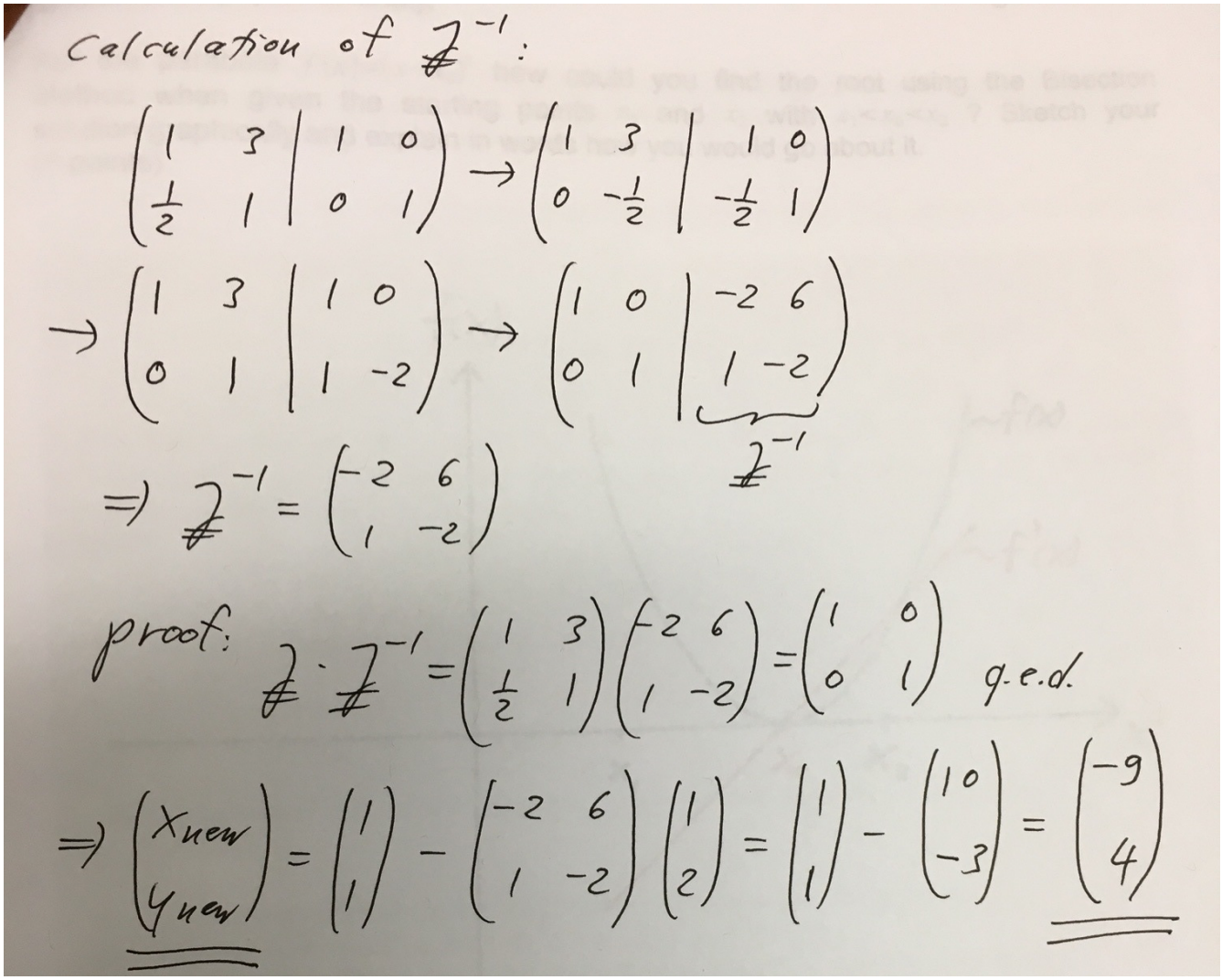
$$\mathcal{J} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} = \begin{pmatrix} y^3 & 3xy^2 \\ \frac{1}{2}x^{-\frac{1}{2}} & 1 \end{pmatrix}$$

it is:

$$\vec{x}_{new} = \vec{x}_{old} + \delta\vec{x} = \vec{x}_{old} - \mathcal{J}^{-1}\vec{F}(\vec{x}_{old}) \quad \text{with } \vec{F}(\vec{x}) = \begin{pmatrix} f(\vec{x}) \\ g(\vec{x}) \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_{new} \\ y_{new} \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \underbrace{\begin{pmatrix} 1 & 3 \\ \frac{1}{2} & 1 \end{pmatrix}^{-1}}_{\text{need to use Gauss-Jordan Elimination to calculate } \mathcal{J}^{-1}} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \text{and } \vec{x}_{old} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

→ next page!



Note: there are several other ways of inverting the matrix J, not just Gauss-Jordan Elimination used above.

Question #9 (6 points total):

a) You have the following parameter vector in the population of a Genetic Algorithm:
$[8 \quad -5 \quad -1 \quad 9 \quad 4 \quad 3 \quad -7 \quad 1]$
Apply the *inversion* operator and write down the resulting parameter vector.
(2 points)

b) The following two parents have been selected for a *crossover* operation in a Genetic Algorithm:

Parent 1: $[1 \quad 3 \quad -1 \quad 9 \quad 4 \quad -5 \quad -7 \quad 8]$

Parent 2: $[8 \quad -2 \quad 0 \quad -1 \quad 6 \quad 3 \quad 9 \quad 0]$

The crossover area starts at position 2 and ends at position 5 including. Which two resulting offspring parameter vectors will result?
(4 points)

a)
$[1 \quad -7 \quad 3 \quad 4 \quad 9 \quad -1 \quad -5 \quad 8]$

b)
Child #1: $[1 \quad -2 \quad 0 \quad -1 \quad 6 \quad -5 \quad -7 \quad 8]$

Child #2: $[8 \quad 3 \quad -1 \quad 9 \quad 4 \quad 3 \quad 9 \quad 0]$

Question #10 (4 points total):

a) Which of the following root finding methods are bracketing the root at all times: 1D Newton-Raphson, Regula Falsi, Secant Method, Bisection Method?
(2 points)

b) What distinguishes the Newton-Raphson Method from the other 3 root finding methods listed in a)? List 2 main differences.
(2 points)

a) Regula Falsi and Bisection Method

b) – Newton-Raphson Method uses explicitly f'(x)
– Newton-Raphson Method uses only one point (x, f(x)) at a time